

- use MITS

switched to db MITS

use MITS

already on db MITS

```
db.students.insertMany([
```

```
{student_id:1,
```

```
name:"Saran",
```

```
department:"CSE",
```

```
city:"Bangalore",
```

```
cgpa:8.5,
```

```
age:21,
```

```
gender:"Male"},
```

```
{student_id:2,
```

```
name:"Priya",
```

```
department:"ECE",
```

```
city:"Hyderabad",
```

```
cgpa:9.0,
```

```
age:22,
```

```
gender:"Female"},
```

```
{student_id:3,
```

```
name:"Rahul",
```

```
department:"CSE",
```

```
city:"Chennai",
```

```
cgpa:7.8,
```

```
age:21,
```

```
gender:"Male"},
```

```
{student_id:4,
```

```
name:"Deepa",
```

```
department:"EEE",
```

city:"Bangalore",
cgpa:8.9,
age:22,
gender:"Female"},
{student_id:5,
name:"Anjali",
department:"ECE",
city:"Pune",
cgpa:8.2,
age:20,
gender:"Female"},
{student_id:6,
name:"Kumar",
department:"Mechanical",
city:"Hyderabad",
cgpa:7.5,
age:23,
gender:"Male"},
{student_id:7,
name:"Sneha",
department:"CSE",
city:"Bangalore",
cgpa:9.1,
age:21,
gender:"Female"},
{student_id:8,
name:"Varun",
department:"Civil",
city:"Chennai",

```
cgpa:8.0,
age:22,
gender:"Male"},
{student_id:9,
name:"Charan",
department:"Mechanical",
city:"Pune",
cgpa:7.2,
age:23,
gender:"Male"},
{student_id:10,
name:"Harsha",
department:"CSE",
city:"Hyderabad",
cgpa:8.7,
age:21,
gender:"Male"}])
```

- db.students.aggregate([
 { \$group: {
 _id: null,
 totalStudents: { \$sum: 1 } } }])

- db.students.aggregate([
 { \$group: {
 _id: "\$department",
 count: { \$sum: 1 } } }])

- db.students.aggregate([
 { \$group: {
 _id: "\$city",
 count: { \$sum: 1 } } }])

- `db.students.aggregate([`
`{ $match: { gender: "Male" } },`
`{ $group: {`
`_id: null,`
`count: { $sum: 1 } } ]])`
- `db.students.aggregate([`
`{ $match: { gender: "Female" } },`
`{ $group: {`
`_id: null,`
`count: { $sum: 1 } } ]])`
- `db.students.aggregate([`
`{ $group: {`
`_id: null,`
`averageCGPA: { $avg: "$cgpa" } } ]])`
- `db.students.aggregate([`
`{ $group: {`
`_id: "$department",`
`averageCGPA: { $avg: "$cgpa" } } ]])`
- `db.students.aggregate([`
`{ $group: {`
`_id: "$department",`
`averageAge: { $avg: "$age" } } ]])`
- `db.students.aggregate([`
`{ $group: {`
`_id: "$city",`
`averageCGPA: { $avg: "$cgpa" } } ]])`
- `db.students.aggregate([`
`{ $group: {`
`_id: null,`

```
averageAge: { $avg: "$age" } } ]])
```

- db.students.aggregate([

```
{ $group: {
```

```
  _id: null,
```

```
  totalCGPA: { $sum: "$cgpa" } } ]])
```

- db.students.aggregate([

```
{ $group: {
```

```
  _id: null,
```

```
  totalAge: { $sum: "$age" } } ]])
```

- db.students.aggregate([

```
{ $group: {
```

```
  _id: "$department",
```

```
  totalCGPA: { $sum: "$cgpa" } } ]])
```

- db.students.aggregate([

```
{ $group: {
```

```
  _id: "$city",
```

```
  totalAge: { $sum: "$age" } } ]])
```

- db.students.aggregate([

```
{ $match: { department: "CSE" } },
```

```
{ $group: {
```

```
  _id: null,
```

```
  totalCGPA: { $sum: "$cgpa" } } ]])
```

- db.students.aggregate([

```
{ $group: {
```

```
  _id: null,
```

```
  highestCGPA: { $max: "$cgpa" } } ]])
```

- db.students.aggregate([

```
{ $group: {
```

```
  _id: null,
```

highestAge: { \$max: "\$age" } } }])

- db.students.aggregate([

{ \$group: {

_id: "\$department",

highestCGPA: { \$max: "\$cgpa" } }]])

- db.students.aggregate([

{ \$group: {

_id: "\$city",

highestAge: { \$max: "\$age" } } }])

- db.students.aggregate([

{ \$group: {

_id: "\$department",

averageCGPA: { \$avg: "\$cgpa" } } },

{ \$sort: { averageCGPA: -1 } },

{ \$limit: 1 }]])

- db.students.aggregate([

{ \$group: {

_id: null,

minimumCGPA: { \$min: "\$cgpa" } } }])

- db.students.aggregate([

{ \$group: {

_id: null,

minimumAge: { \$min: "\$age" } } }]])

- db.students.aggregate([

{ \$group: {

_id: "\$department",

minimumCGPA: { \$min: "\$cgpa" } } }]])

- db.students.aggregate([

{ \$group: {

```

    _id: "$city",
    minimumAge: { $min: "$age" } } }])

    • db.students.aggregate([
{ $group: {
    _id: "$city",
    averageCGPA: { $avg: "$cgpa" } } },
{ $sort: { averageCGPA: 1 } },
{ $limit: 1 }])

    • db.students.aggregate([
{ $group: {
    _id: "$department",
    count: { $sum: 1 } } }])

    • db.students.aggregate([
{ $group: {
    _id: "$city",
    count: { $sum: 1 } } }])

    • db.students.aggregate([
{ $group: {
    _id: "$gender",
    count: { $sum: 1 } } } ]])

    • db.students.aggregate([
{ $group: {
    _id: "$department",
    averageCGPA: { $avg: "$cgpa" } } }])

    • db.students.aggregate([
{ $group: {
    _id: "$city",
    averageAge: { $avg: "$age" } } }])

    • db.students.aggregate([

```

```

{ $match: { cgpa: { $gt: 8 } } },
{
  $group: {
    _id: null,
    count: { $sum: 1 } } })

• db.students.aggregate([
{ $match: { department: "CSE" } },
{
  $group: {
    _id: null,
    averageCGPA: { $avg: "$cgpa" } } })

• db.students.aggregate([
{ $match: { gender: "Female" } },
{
  $group: {
    _id: null,
    totalFemales: { $sum: 1 } } })

• db.students.aggregate([
{ $match: { gender: "Male" } },
{
  $group: {
    _id: null,
    highestCGPA: { $max: "$cgpa" } } })

• db.students.aggregate([
{ $match: { gender: "Female" } },
{
  $group: {
    _id: null,
    lowestCGPA: { $min: "$cgpa" } } })

• db.students.aggregate([
{
  $group: {
    _id: "$department",
    averageCGPA: { $avg: "$cgpa" } },
{ $sort: { averageCGPA: -1 } })

```


- `db.students.aggregate([`
`{ $group: {`
`_id: "$city",`
`count: { $sum: 1 } }`
`{ $sort: { count: -1 } }])`
- `db.students.aggregate([`
`{ $group: {`
`_id: "$department",`
`maxCGPA: { $max: "$cgpa" } }`
`{ $sort: { maxCGPA: -1 } }])`
- `db.students.aggregate([`
`{ $group: {`
`_id: "$city",`
`averageAge: { $avg: "$age" } }`
`{ $sort: { averageAge: -1 } }])`
- `db.students.aggregate([`
`{ $group: {`
`_id: "$department",`
`count: { $sum: 1 } }`
`{ $sort: { count: -1 } }])`